

Few-Shot Continual Learning for Audio Deepfake Detection in Financial Voice Authentication

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Abstract—Audio deepfakes have become a direct risk to financial voice authentication because modern text-to-speech and voice conversion systems can imitate a customer well enough to challenge automated speaker verification and human call-center review. This paper synthesized a finance-focused literature review, a proposed system design, and an implementation report into a single IEEE-style study of a few-shot continual learning framework for audio deepfake detection. The proposed architecture combines telephone-aware preprocessing, dual-stream feature extraction from WavLM, HuBERT, MFCC, LFCC, and CQCC descriptors, a Prototypical Network embedding space, and Elastic Weight Consolidation for future adaptation to unseen attacks. The implemented system was trained on the ASVspoof 2019 Logical Access dataset using episodic few-shot training over six known spoofing attacks. After seven completed epochs, the model reached 99.17% prototypical training accuracy and 96.85% best development accuracy, while training loss decreased from 1.1295 to 0.0962. These results show that the representation learned by the proposed pipeline separates bonafide and spoofed speech effectively on known attacks. The study also identified the main barrier to financial deployment: current benchmarks still fall short of the noisy, codec-compressed, rapidly changing conditions faced by production banking systems.

Index Terms—audio deepfake detection, financial fraud, voice authentication, few-shot learning, continual learning, prototypical networks

I. INTRODUCTION

Voice has become a practical biometric in banking, fintech support, and telephone-based customer service because it reduces friction during identity verification and fits natural call-center workflows. At the same time, the attack surface has expanded sharply. Recent surveys show that audio deepfake generation has progressed from laboratory demonstrations to realistic impersonation tools that can clone timbre, prosody, and speaking style with limited enrollment audio [1], [2]. In financial settings this threat is not abstract. Vulnerability studies have shown that commercial voice biometric systems can be bypassed by cloned speech, and financial institutions are exposed because many authentication flows still assume that speech arriving through the telephone channel is difficult to forge [3], [4]. A successful spoof can translate into account takeover, fraudulent transfer approval, or manipulation of high-value customer support interactions.

The operational context makes the problem harder than generic synthetic speech detection. Banking calls often contain channel compression, packet loss, mixed devices, variable

background noise, and short utterances. The same call may move between mobile, Voice over IP, and landline conditions. Work on partial fake speech has further shown that an adversary does not need to synthesize an entire interaction; modifying a short span that contains an account number or authorization phrase may be enough to create financial harm [5]. Datasets and evaluations that focus on clean recordings therefore capture only part of the deployment problem. Large benchmark efforts such as ASVspoof 5 and EchoFake have moved the field toward more realistic assessment, yet the literature review that underpins this study found that financial-domain audio remains underrepresented and that telephone-grade robustness is still a weak point for many detectors [6]–[8].

Another gap concerns adaptation. Fraud tools change quickly. New synthesis engines, voice conversion pipelines, codec-aware post-processing, and replay strategies appear much faster than institutions can curate labeled training corpora. A banking countermeasure that needs thousands of examples for every new attack class is not operationally attractive. The reviewed literature identified only limited work on few-shot adaptation to unseen spoofing conditions, despite the fact that few-shot update capability is one of the most natural requirements for fraud response teams [9]–[11]. This challenge becomes sharper when continual learning is considered. Even if a system can adapt to a new generator with a small support set, that adaptation is of limited value if it erodes performance on older attacks that still circulate in fraud campaigns.

Financial authentication also imposes asymmetric decision costs. A false acceptance can authorize unauthorized access or funds movement, whereas a false rejection can delay service, trigger manual escalation, and reduce trust in voice channels. A detector for this domain therefore has to do more than maximize benchmark accuracy. It should produce stable scores that support threshold tuning, and it should admit controlled updates when fraud analysts confirm that a new spoofing tool has entered circulation.

This paper addresses that gap by integrating three artifacts from one research project: a curated literature review focused on financial fraud, a proposed methodology for a few-shot continual detector, and an implementation report with experimental findings. The resulting paper has two goals. The first is conceptual: to present a clear finance-aware formulation

of audio deepfake detection in which data scarcity, channel mismatch, and catastrophic forgetting are treated as first-class constraints, not as afterthoughts. The second is empirical: to document how the proposed design was instantiated on the ASVspoof 2019 Logical Access benchmark and what the first round of experiments established about representation quality, convergence, and development-set generalization.

The work makes four contributions.

- 1) It synthesizes 24 recent studies on audio deepfake detection into a financial threat model that emphasizes telephone channels, short utterances, unseen generators, and operational update requirements.
- 2) It formulates audio deepfake detection for financial voice authentication as a few-shot continual learning problem with explicit support and query structure, prototype-based inference, and forgetting-aware adaptation constraints.
- 3) It proposes a dual-stream detection architecture that combines self-supervised speech representations with cepstral descriptors and couples a Prototypical Network backbone with Elastic Weight Consolidation.
- 4) It reports the implementation outcome on ASVspoof 2019 LA, where the system reached 96.85% best development prototypical accuracy after seven epochs and exhibited rapid loss reduction from 1.1295 to 0.0962.

The contribution is intentionally bounded. The current implementation validates the learned representation on known attacks in the training and development partitions. It does not yet provide a completed held-out few-shot adaptation study over attacks A07-A19. That limitation matters for interpretation, so it is treated explicitly in the experimental and limitations sections instead of being hidden behind broad claims.

The remainder of the paper is organized as follows. Section II reviews the literature thematically and identifies the main technical and deployment gaps. Section III formalizes the problem. Section IV presents the proposed methodology and system design. Section V describes the implementation details that instantiate that design. Section VI reports the experimental setup and quantitative results. Section VII interprets the findings. Section VIII states the limitations. Section IX concludes the paper, and Section X outlines concrete next steps.

II. LITERATURE REVIEW

The reviewed literature converges on one central point: audio deepfake detection is no longer a narrow speech-processing task. In financial settings it is an adversarial security problem in which data, channel conditions, model design, and update strategy interact. For that reason, a thematic synthesis is more useful than a chronological list of papers.

The first theme is the financial threat landscape. General surveys now position voice deepfakes as a mature social engineering and authentication risk, not a speculative concern [1], [2]. Delgado et al. framed the problem around telephone banking and argued that presentation conditions shape detector performance as much as the synthesis model itself [4]. Hong et al. extended that concern by benchmarking cloned

speech against commercial biometric systems and showing that production voice authentication can be substantially more vulnerable than controlled academic evaluations suggest [3]. Alali and Theodorakopoulos added an important refinement by studying partial fake speech attacks, which are especially relevant to finance because a brief manipulated phrase may be more dangerous than a fully synthetic conversation [5]. Banking fraud studies and multimodal financial AI work further suggest that audio evidence should be fused with other risk indicators instead of being treated as a standalone oracle [11], [12].

The second theme concerns detector architectures. End-to-end deep learning models have displaced purely handcrafted pipelines because they learn richer attack cues and adapt better to varied acoustic conditions. DeepLASD is representative of this movement: it learns directly from audio patterns relevant to logical access spoofing and targets real-time countermeasure use cases [13]. Gong and Li pushed this line further with a Transformer architecture that uses cross-attention to fuse acoustic representations, showing that fusion is not only a feature engineering choice but also an architectural one [14]. Hybrid architectures remain common because they combine complementary inductive biases. AudioFakeNet joins convolutional, recurrent, and attention modules to capture local spectral structure, temporal evolution, and salient regions in one model [15]. RPRA-ADD instead emphasizes explicit forgery traces, a useful direction for security settings that need localized evidence [16]. Across these studies, strong detectors tend to combine multiple signal views through feature fusion or multi-branch design.

A third theme is the role of datasets and evaluation frameworks. Benchmark progress has been significant, but the literature also shows why benchmark success does not guarantee deployment readiness. ASVspoof remains the reference family of corpora for anti-spoofing research, and ASVspoof 5 broadened that role by collecting crowdsourced speech and publishing a large challenge evaluation [6], [7]. The evaluation paper is particularly informative because it documents that codec processing and adversarial enhancement still cause notable degradation. EchoFake responded to another weak spot by emphasizing replay-aware and device-diverse conditions that are closer to practical telephony [8]. SONAR then made the generalization problem harder to ignore by evaluating multiple detectors against advanced generators and showing that cross-generator robustness is often much weaker than within-distribution performance [21]. FinAudio is not a deepfake benchmark, but it is still relevant because it marks the emergence of finance-specific audio resources and underlines how domain mismatch affects speech systems in financial applications [10]. Taken together, these studies show that evaluation should not stop at clean train-test splits. It must probe channel variation, unseen generators, and application-specific speech.

The fourth theme is feature learning. Handcrafted descriptors remain useful because they capture spectral artifacts that many neural encoders do not isolate cleanly. The multicepstral

analysis study showed that MFCC, LFCC, and CQCC combinations can remain competitive, especially when computation and latency matter [18]. At the same time, self-supervised learning has changed the field by offering representations trained on large unlabeled speech corpora. The improved NeXt-TDNN study demonstrated that fusing WavLM and HuBERT features can improve robustness compared with single-source embeddings [17]. Beyond Identity and AFSS extend this idea in complementary ways. The first argues that detectors should separate deepfake artifacts from speaker identity so that success does not depend on knowing the speaker well in advance [20]. The second addresses bias by generating artifact-focused self-synthesis examples that reduce identity leakage during training [19]. This literature strongly supports a dual-stream design in which self-supervised and cepstral features are treated as complementary and not as competing representations.

The fifth theme is generalization and adaptation. The few-shot adaptation study reviewed in this project is notable because it tackles the exact setting that fraud response teams face: a new spoofing tool appears, only a small support set is available, and the detector must adapt under distribution shift [9]. Survey work reinforces the same conclusion from a broader angle. The attack-perspective review, the systematic survey literature, and the ACM Computing Surveys article all argue that unseen attack families, channel variability, and adversarial evolution are persistent gaps, not short-term anomalies [2], [22], [23]. These papers do not merely call for better classifiers. They point to a different learning problem in which adaptation and retention must be optimized together.

Table I summarizes representative papers that shaped the design logic of the present framework. The selected studies cover finance-specific threat analysis, architectural trends, evaluation resources, and few-shot generalization.

TABLE I
REPRESENTATIVE LITERATURE COMPARISON

Author/Year	Method	Dataset	Key Metric	Limitation
Delgado et al., 2025	Presentation-aware deepfake voice detection for telephone banking	Real-world banking conditions	Detection score calibration	Strongly tied to channel presentation; limited evidence on continual adaptation
Al-Tairi et al., 2025	DeepLAsD end-to-end countermeasure	Logical access spoofing benchmarks	Detection accuracy	Focuses on supervised detection instead of unseen attack adaptation
Gong and Li, 2025	Cross-attention Transformer with acoustic feature fusion	ASVspoof 2019 LA	EER and accuracy	Architecture is effective, but channel mismatch remains a concern
Wang et al., 2025	ASVspoof benchmark design	5 Crowdsourced speech corpus	Challenge evaluation metrics	Benchmark breadth improved, but finance-specific dialogue is absent
Zhang et al., 2025	EchoFake replay-aware corpus	Telephone and device-diverse recordings	Real-world replay evaluation	Replay emphasis does not fully cover continual spoof adaptation
Authors, 2025	Few-shot spoof adaptation under distribution shift	Unseen synthetic speech attacks	Few-shot detection performance	Does not address the financial deployment setting directly
Authors, 2025	Improved NeXt-TDNN with fused SSL features	ASVspoof benchmarks	Robustness across benchmarks	High feature quality comes with notable compute cost
Ahmadiadi et al., 2025	Identity-independent artifact detection	Cross-generator evaluation settings	Generalization performance	Requires careful disentanglement between speaker and artifact cues

The gap that emerges from this synthesis is narrow enough to define clearly. Existing work has improved detector accuracy, richer features, and evaluation realism. Even so, the finan-

cial setting still lacks a framework that combines telephone-aware preprocessing, complementary feature fusion, few-shot update capability, and protection against catastrophic forgetting in a single operational design. That gap motivates the problem definition and methodology in the following sections.

III. PROBLEM STATEMENT

Consider a financial voice authentication system that receives an input utterance $x \in \mathbb{R}^T$ and must decide whether the speech is bonafide or spoofed. Let the binary label be $y \in \{0, 1\}$, where $y = 0$ denotes bonafide speech and $y = 1$ denotes spoofed speech. Each spoofed sample may also carry an attack identifier $a \in \mathcal{A}$, where $\mathcal{A}_{\text{train}} = \{A01, \dots, A06\}$ contains attack types observed during training and $\mathcal{A}_{\text{novel}} = \{A07, \dots, A19\}$ contains unseen attacks reserved for later adaptation and evaluation.

During initial training, the model has access to a dataset

$$\mathcal{D}_{\text{train}} = \{(x_i, y_i, a_i)\}_{i=1}^N, \quad a_i \in \mathcal{A}_{\text{train}} \cup \{\text{bonafide}\}. \quad (1)$$

The first objective is to learn a detector f_θ that maps an utterance to a decision score while preserving separation between bonafide and spoofed speech across speakers, sessions, and channel conditions:

$$f_\theta : x \mapsto \hat{y}, \quad \hat{y} \in \{0, 1\}. \quad (2)$$

In a few-shot adaptation setting, a previously unseen attack $a^* \in \mathcal{A}_{\text{novel}}$ appears with only a small support set

$$\mathcal{S}^{a^*} = \{(x_j, y_j = 1, a_j = a^*)\}_{j=1}^K, \quad K \in \{5, 10, 20\}. \quad (3)$$

The system must update its representation or decision boundary using this limited evidence while retaining performance on the known attacks already learned. This leads to a joint requirement:

$$\min_{\theta'} \mathcal{L}_{\text{novel}}(\theta'; \mathcal{S}^{a^*}, \mathcal{Q}^{a^*}) + \lambda \sum_m F_m(\theta'_m - \theta_m^*)^2, \quad (4)$$

where $\mathcal{Q}^{a^*}$ is a query set for the novel attack, θ^* are parameters learned on known attacks, and F_m are Fisher-weighted importance terms used to penalize drift in parameters that matter for previously learned tasks.

The initialization stage is itself episodic. Let $\mathcal{E} = \{(\mathcal{S}_e, \mathcal{Q}_e)\}_{e=1}^M$ denote a sequence of few-shot episodes, where each support set and query set contains balanced bonafide and spoof samples. The representation is first learned by minimizing the average episodic objective

$$\min_{\theta} \frac{1}{M} \sum_{e=1}^M \left(\mathcal{L}_{\text{proto}}^{(e)} + \mathcal{L}_{\text{cls}}^{(e)} + \beta \mathcal{L}_{\text{atk}}^{(e)} \right), \quad (5)$$

where β controls the auxiliary attack-type supervision term. This formulation matters because the downstream use case is also episodic: the deployed model will compare a small query batch against a small support set when a new fraud pattern is confirmed.

Deployment decisions are thresholded scores, not raw labels alone. If $s_\theta(x)$ is the spoof score produced by the detector and τ is a decision threshold, then the operating characteristics are

$$\text{FAR}(\tau) = \frac{|\{x \in \mathcal{D}_{\text{bonafide}} : s_\theta(x) \geq \tau\}|}{|\mathcal{D}_{\text{bonafide}}|}, \quad (6)$$

$$\text{FRR}(\tau) = \frac{|\{x \in \mathcal{D}_{\text{spoof}} : s_\theta(x) < \tau\}|}{|\mathcal{D}_{\text{spoof}}|}. \quad (7)$$

The Equal Error Rate is defined at the threshold τ^* for which $\text{FAR}(\tau^*) = \text{FRR}(\tau^*)$. These quantities matter operationally because they formalize the trade-off between fraud pass-through and customer friction.

The problem can therefore be stated formally as follows. Given channel-degraded speech segments, known spoofing attacks, and only a few labeled examples from new attacks, learn an embedding and decision rule that satisfy three properties: high bonafide-versus-spoof discrimination on known conditions, rapid adaptation to novel spoofing conditions, and bounded forgetting on previously learned attacks. In the financial context, this problem is constrained by latency, by the high cost of both false accepts and false rejects, and by the fact that attack evolution is continuous, not episodic.

Retention must also be measurable across adaptation stages. If $R_{i,i}$ denotes performance on task i immediately after learning it and $R_{T,i}$ denotes performance on that same task after the final adaptation stage T , then backward transfer is

$$\text{BWT} = \frac{1}{T-1} \sum_{i=1}^{T-1} (R_{T,i} - R_{i,i}). \quad (8)$$

Negative values indicate forgetting. This metric is central in the present setting because a financial countermeasure that gains one new attack class while losing older ones does not improve real fraud resilience.

The practical implication is that conventional closed-set anti-spoofing is insufficient. A detector optimized only for static train-test splits can perform well on benchmark partitions yet fail in production when the fraudster changes the generator or when the call path changes the acoustic presentation. The goal of the present framework is to move from that closed-set assumption toward a representation that remains useful when the spoofing surface changes.

IV. PROPOSED METHODOLOGY

The proposed methodology combines ideas from the literature review with the project's system design document. It is organized around five components: data strategy, telephone-aware preprocessing, dual-stream feature extraction, prototype-based detection, and continual adaptation. The design is motivated by the observation that finance-grade deployment requires more than raw classification accuracy. It also requires resilience to channel effects, compact adaptation updates, and an explicit mechanism for retaining earlier knowledge.

A. Design Objectives

The framework was designed to satisfy four technical objectives. First, it should learn a representation that separates bonafide and spoofed speech even when the spoof is generated by a tool that differs from the tools seen during training. Second, it should admit few-shot updates from a small support set instead of requiring a full retraining cycle. Third, it should reduce forgetting after those updates. Fourth, it should remain consistent with the constraints of telephone banking, where the audio path is lossy and the interaction often contains only short decision-critical utterances. These objectives shape the choice of preprocessing, features, model structure, and adaptation rule.

B. Data Strategy

The conceptual data strategy described in the proposal is broader than the realized implementation. The proposal calls for genuine speech from telephone-quality corpora and fake speech generated by multiple text-to-speech and voice conversion systems, with some synthesis systems held out completely until evaluation. ASVspoof 5 and EchoFake were identified as supporting benchmarks for cross-dataset and replay-aware assessment [6], [8]. This multi-source design is appropriate because it exposes the model to diverse channel and generator conditions.

The current implementation instantiated the design on ASVspoof 2019 LA, which still supports the core few-shot logic because it separates six known training attacks from thirteen unseen evaluation attacks. That choice traded domain breadth for a controlled benchmark with established protocols. The methodology in this paper therefore reflects both levels of the project: the proposed broader deployment-oriented data strategy and the implemented benchmark-centered realization.

C. Telephone-Aware Preprocessing

The preprocessing pipeline transforms raw utterances into segments that are more consistent with financial telephony and more suitable for feature extraction. All audio is resampled to 16 kHz and peak-normalized to the interval $[-1, 1]$. Resampling removes sampling-rate mismatch, while normalization reduces amplitude variability across devices and recordings. These are simple operations, but they prevent the downstream representation from wasting capacity on nuisance variation.

Codec simulation is the next design choice. The proposal includes G.711 for public switched telephone network conditions, AMR-NB for cellular compression, and Opus for Voice over IP conditions. The implementation applies one of these simulated degradations during training with a probability of 50%. This decision follows directly from the literature gap on presentation mismatch: a detector trained only on clean audio is unlikely to retain performance when speech is compressed, bandwidth-limited, or mildly distorted [4], [7]. By injecting codec effects into training, the model is encouraged to focus on spoofing artifacts that survive transmission and away from fragile cues that disappear after compression.

Voice activity detection removes silence regions, and a sliding window extracts fixed-length 3-second segments with 1.5-second overlap. The choice of fixed windows standardizes the input and increases the number of training instances derived from each utterance. The overlap reduces the chance that a discriminative cue is split across segment boundaries. In financial call handling this matters because the relevant evidence may occur in a brief phrase and not across an entire conversation.

D. Dual-Stream Feature Extraction

The feature layer is the methodological core of the system. The first stream uses self-supervised speech encoders. WavLM and HuBERT were chosen because both models learn broad acoustic structure from large unlabeled corpora and have shown strong downstream performance in speech tasks [24], [25]. For each segment, mean-pooled WavLM and HuBERT embeddings are concatenated to produce a 1,536-dimensional self-supervised descriptor. This stream is intended to capture high-level regularities in speech production and synthesis.

The second stream uses cepstral descriptors: MFCC, LFCC, and CQCC. Each contributes 40 dimensions after temporal aggregation, for a total of 120 dimensions. The motivation is not nostalgia for older features. It is complementarity. The literature shows that cepstral features still expose spectral irregularities produced by vocoders, resynthesis pipelines, and frequency-domain artifacts that a generic neural embedding may smooth over [17], [18]. The fused representation is therefore

$$\mathbf{z} = [\mathbf{z}_{\text{ssl}}; \mathbf{z}_{\text{cep}}] \in \mathbb{R}^{1656}, \quad (9)$$

where $\mathbf{z}_{\text{ssl}} \in \mathbb{R}^{1536}$ and $\mathbf{z}_{\text{cep}} \in \mathbb{R}^{120}$.

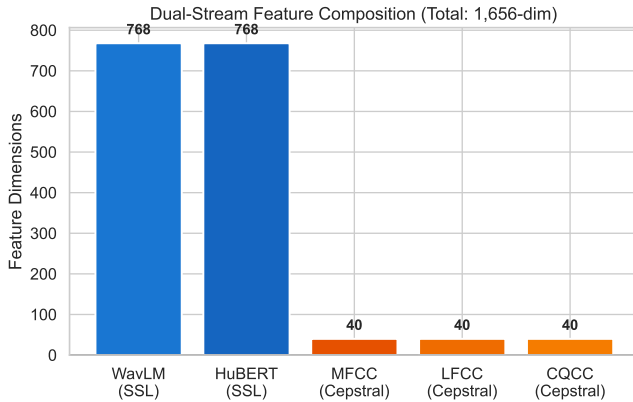


Fig. 1. Dual-stream feature composition. WavLM and HuBERT contribute 1,536 dimensions, while MFCC, LFCC, and CQCC contribute 120 dimensions, for a total fused representation of 1,656 dimensions.

Fig. 1 makes the allocation of representational capacity explicit. Most dimensions come from the self-supervised stream, which captures broad acoustic context, while the cepstral stream remains compact and targeted. This balance is deliberate: the design relies on SSL features for general speech structure and on cepstral descriptors for attack artifacts

that are easy to lose when only one representation family is used.

Placeholder for the system architecture diagram. The final figure should depict raw audio ingestion, preprocessing blocks for resampling, normalization, codec simulation, VAD, and segmentation, a dual-stream feature extractor with WavLM, HuBERT, MFCC, LFCC, and CQCC branches, a Prototypical Network backbone with binary and attack-type heads, an Elastic Weight Consolidation module, and an inference path that returns a deepfake score and supports later few-shot adaptation.

Fig. 2. System architecture of the proposed few-shot continual learning framework for financial voice authentication.

E. Prototype-Based Detection Backbone

The fused feature vector is mapped to a 256-dimensional embedding through a multilayer perceptron. The representation is trained with a Prototypical Network objective because prototype-based inference matches the few-shot requirement naturally [26]. Given a support set $\mathcal{S}_c$ for class c , the class prototype is

$$\mathbf{p}_c = \frac{1}{|\mathcal{S}_c|} \sum_{(x_i, y_i) \in \mathcal{S}_c} g_\theta(x_i), \quad (10)$$

where $g_\theta(\cdot)$ is the embedding function. A query example x_q is classified by distances to the prototypes:

$$P(y = c | x_q) = \frac{\exp(-d(g_\theta(x_q), \mathbf{p}_c))}{\sum_{c'} \exp(-d(g_\theta(x_q), \mathbf{p}_{c'}))}, \quad (11)$$

with $d(\cdot, \cdot)$ taken as squared Euclidean distance. This inference rule is attractive for two reasons. It supports low-shot updates by recomputing or adjusting prototypes from a small support set, and it produces scores with a geometric interpretation that can be calibrated for risk thresholds.

The architecture also includes a binary classification head and an auxiliary attack-type classifier. The binary head preserves a direct bonafide-versus-spoof objective even when the prototype structure becomes noisy. The attack-type head encourages the embedding space to capture finer-grained spoofing structure instead of collapsing all fake speech into one diffuse cluster. That auxiliary objective is useful because different generators may leave distinct artifacts, and retaining that structure should help later adaptation.

F. Continual Adaptation with EWC

Prototype updates alone may be insufficient when a novel attack has a distribution that differs markedly from the training attacks. For this reason, the proposed methodology adds Elastic Weight Consolidation [27]. After training on known attacks, the model estimates parameter importance through the diagonal Fisher information matrix. During later adaptation, changes to important parameters are penalized, which reduces the risk that learning a new attack degrades performance on older attacks.

The combined training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{proto}} + \mathcal{L}_{\text{cls}} + 0.5\mathcal{L}_{\text{atk}} + \lambda_{\text{ewc}}\mathcal{L}_{\text{ewc}}, \quad (12)$$

where $\mathcal{L}_{\text{proto}}$ is the prototypical loss, $\mathcal{L}_{\text{cls}}$ is the binary classification loss, $\mathcal{L}_{\text{atk}}$ is the auxiliary attack-type loss, and $\mathcal{L}_{\text{ewc}}$ is activated during adaptation. In the implemented system, $\lambda_{\text{ewc}} = 5000$.

G. Inference and Adaptation Workflow

In deployment, an incoming segment moves through pre-processing and feature extraction, receives an embedding, and is scored against the current decision rule. If the score exceeds a threshold, the system can flag the utterance for additional verification or human review. If investigators later confirm that the event corresponds to a new spoofing tool, the support examples from that tool can be used to update class prototypes and, if necessary, to fine-tune the embedding under the EWC constraint. This creates a feedback loop in which fraud response and model adaptation are linked and not treated as separate processes.

Placeholder for the workflow diagram. The final figure should show the sequence raw audio input $\rightarrow$ pre-processing $\rightarrow$ dual-stream feature extraction $\rightarrow$ embedding and scoring $\rightarrow$ decision threshold. A secondary feedback branch should depict the arrival of a confirmed novel attack, creation of a small support set, prototype update, EWC-regularized fine-tuning, and reinsertion of the adapted model into the inference loop.

Fig. 3. End-to-end workflow of the proposed detection and adaptation pipeline.

The methodological choice to combine prototype learning with EWC is therefore not arbitrary. Prototype learning provides an efficient way to exploit small support sets. EWC provides a retention mechanism. The fused feature stack improves the odds that the underlying embedding contains both high-level speech structure and lower-level spoofing artifacts. Together, these components address the central gap identified in the literature review.

V. IMPLEMENTATION DETAILS

The implemented system realized the methodology as a modular Python pipeline on Ubuntu Linux with Python 3.11, PyTorch 2.x, Hugging Face Transformers, torchaudio, librosa, SciPy, scikit-learn, Pandas, NumPy, WebRTC VAD, Matplotlib, and Seaborn. Training ran on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. This environment was chosen because dual self-supervised encoders are expensive to execute repeatedly during episodic training, and GPU acceleration is necessary to keep experimentation practical.

A. Software Architecture

The implementation is organized into six logical modules. A configuration module stores audio, model, training, and inference hyperparameters. A preprocessing module handles resampling, normalization, codec simulation, VAD, segmentation, and dataset construction. A feature module computes the two feature streams. A model module defines the embedding network, prototypical head, binary head, auxiliary attack classifier, and EWC logic. A training module implements episodic optimization and checkpointing. An inference module exposes threshold-based scoring and few-shot adaptation utilities. This modular split mirrors the system design and makes it easier to isolate the cost and effect of each stage.

B. Dataset Realization

Although the broader proposal envisioned multi-source data, the implementation focused on the ASVspoof 2019 LA corpus. This dataset remains suitable because it includes disjoint training, development, and evaluation partitions, each with fixed speaker splits and explicit attack identifiers. Table II summarizes the characteristics that matter most for the implemented study.

TABLE II
ASVspoof 2019 LA CHARACTERISTICS USED IN THE IMPLEMENTATION

Property	Value
Audio format	16-bit FLAC, mono, 16 kHz
Training samples	25,380
Development samples	24,844
Training speakers	20 (8 male, 12 female)
Development speakers	10 (4 male, 6 female)
Known training attacks	A01-A06
Held-out evaluation attacks	A07-A19
Bonafide class	Genuine speech
Spoof class	TTS and VC speech

The implementation preserved the challenge’s disjoint partition structure. No speaker overlap exists between train and development, which is useful because it limits the chance that the detector inflates performance by memorizing speaker identity rather than learning spoofing artifacts. The evaluation partition was reserved conceptually for later few-shot adaptation experiments, consistent with the project design.

The partitioning strategy also shaped the experimental claims that could be made safely. Because the development set contains unseen speakers but known attack families, it tests whether the learned embedding captures spoofing evidence that transfers across speakers. It does not test whether the same embedding transfers across novel synthesis systems. That second question belongs to the held-out evaluation partition and explains why the present paper treats representation learning and adaptation as separate empirical stages.

C. Feature and Model Configuration

The realized feature stack matches the proposal closely. WavLM-base-plus and HuBERT-base-ls960 are loaded as frozen feature extractors, yielding 768 dimensions each after

mean pooling. MFCC, LFCC, and CQCC contribute 40 dimensions each. The resulting 1,656-dimensional vector enters a three-layer multilayer perceptron with batch normalization and dropout of 0.3, and the output embedding dimension is 256. The model further includes a binary classifier for bonafide-versus-spoof prediction and an auxiliary multiclass head for attack-type supervision. The total number of trainable parameters in the learned detector is 1,311,369, excluding the frozen self-supervised encoders.

The implementation choice to freeze WavLM and HuBERT matters. It reduces training instability and constrains the number of trainable parameters, but it also shifts computational burden into repeated forward passes through two large encoders. As the results later show, this choice contributes substantially to the per-epoch runtime.

D. Training Configuration

Training followed an episodic few-shot protocol in which each episode was a 2-way classification problem with 5 support examples per class and 15 query examples per class. Each epoch comprised 1,000 training episodes and 200 development episodes. The optimizer was Adam with learning rate 1×10^{-4} and weight decay 1×10^{-5} . A StepLR scheduler halved the learning rate every 20 epochs. Gradient clipping used a maximum norm of 1.0, early stopping patience was set to 10 epochs, and the EWC regularization coefficient was 5000. Table III summarizes the realized configuration.

TABLE III
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Training paradigm	Episodic few-shot
N -way	2
K -shot support	5 per class
Query examples	15 per class
Episodes per epoch	1,000
Planned epochs	50
Completed epochs	7
Optimizer	Adam
Learning rate	1×10^{-4}
Weight decay	1×10^{-5}
Scheduler	StepLR, step 20, gamma 0.5
Dropout	0.3
Gradient clipping	1.0
Attack-type loss weight	0.5
EWC coefficient	5000
Random seed	42

The implementation report planned for 50 epochs, but only seven had completed when the results were documented. That fact affects how the quantitative findings should be read. They are evidence of representation learning quality and convergence behavior, not the final ceiling of the system.

E. Execution Flow and Resource Profile

The execution path of one training epoch helps explain both the strengths and the costs of the implemented system. Each episode begins with protocol-driven sampling of bonafide and spoof examples, followed by waveform preprocessing, segmentation, and dual-stream feature extraction. Only after those

front-end steps does the compact trainable network operate on the fused 1,656-dimensional vector. The consequence is a model with a modest number of trainable parameters but a heavy front end. The detector itself is small enough to optimize comfortably on a single GPU, yet the repeated forward passes through WavLM and HuBERT dominate wall-clock time.

The development loop mirrors the training loop but disables gradient updates and evaluates only 200 sampled episodes. This keeps validation cheaper than training while preserving the episodic structure needed for a meaningful few-shot estimate. The implementation report also makes clear that checkpointing followed development loss instead of training accuracy. New best checkpoints were saved at epochs 1, 2, and 6. That policy is appropriate because the project is interested in transferable representations, and development loss is a closer proxy for that objective than raw training accuracy alone.

From a systems perspective, the implementation is therefore a deliberate trade-off. Frozen self-supervised encoders stabilize optimization and inject strong prior speech knowledge, but they shift the runtime burden to feature computation. The prototype-based back end and auxiliary heads remain lightweight. This asymmetry explains why the model can converge quickly in only a few epochs while still requiring substantial wall-clock time per epoch.

VI. EXPERIMENTS AND RESULTS

The experimental study was designed to answer a restricted but useful question: can the implemented dual-stream prototype-based system learn a stable bonafide-versus-spoof representation on the known ASVspoof 2019 LA attacks, and how quickly does that learning occur? Because the reported implementation had not yet completed held-out attack adaptation, the present section focuses on known-attack training and development behavior.

A. Experimental Setup

Training used the ASVspoof 2019 LA training split with attacks A01-A06 and bonafide speech. Development evaluation used the official development split with disjoint speakers. Each development pass sampled 200 few-shot episodes, which means the measured development loss contains some variance from episode composition in addition to model quality. The evaluation partition containing attacks A07-A19 was kept outside the reported results. That choice is consistent with the original project plan, which reserved it for later few-shot adaptation and backward-transfer analysis.

B. Evaluation Criteria

The evaluation design follows directly from the project methodology. Equal Error Rate, accuracy, precision, recall, F1 score, prototypical classification accuracy, and Backward Transfer were identified as the complete metric set for the framework. In the seven-epoch implementation report, the emphasis fell on prototypical classification accuracy and loss because those metrics answer the first experimental question cleanly: whether the learned embedding is organizing bonafide

and spoofed speech correctly on sampled few-shot episodes before any held-out adaptation takes place.

This distinction is important. Prototypical accuracy measures the percentage of query samples that are assigned to the correct class after prototype construction inside each episode. It therefore tracks exactly the inference rule that the system would use in low-shot settings. Training and development loss measure convergence of the joint objective in (12). EER, precision, recall, and F1 remain essential for later deployment-oriented evaluation because they capture threshold behavior and class-specific error costs. Backward Transfer is also essential, but only after the adaptation stage is executed. By separating these questions, the current experiments avoid claiming continual learning evidence that the seven-epoch implementation had not yet produced.

The metrics tracked in the design document were Equal Error Rate, accuracy, precision, recall, F1 score, prototypical classification accuracy, and Backward Transfer. For the implemented training phase, the primary reported metrics were prototypical classification accuracy and loss on the training and development episodes because they directly reflect optimization progress before adaptation. EER, precision, recall, and Backward Transfer remain part of the full framework, but they were not the main reported numbers in the implementation document because held-out adaptation had not yet been completed.

Baselines were limited in the empirical report. No internal reimplementations of a traditional cepstral-only or end-to-end comparison model was included in the seven-epoch run. As a result, the experiments used two comparison anchors instead. The first anchor was the training-versus-development gap across checkpoints. The second anchor was the literature reviewed earlier, particularly handcrafted multicepstral systems, fused SSL systems, and few-shot spoofing studies [9], [17], [18]. This is less rigorous than a controlled head-to-head baseline table, but it is the honest scope of the available implementation evidence.

C. Per-Epoch Performance

Table IV presents the reported metrics for the seven completed epochs. The training loss decreased monotonically from 1.1295 to 0.0962, while the training prototypical accuracy rose from 88.10% to 99.17%. The development accuracy peaked at 96.85% in epoch 6, where the development loss reached its minimum value of 0.3097.

TABLE IV
PER-EPOCH TRAINING AND DEVELOPMENT METRICS

Epoch	Train Loss	Train Acc.	Dev Loss	Dev Acc.
1	1.1295	88.10%	0.8702	92.50%
2	0.2883	98.02%	0.4220	94.20%
3	0.1569	98.73%	0.4383	94.65%
4	0.1279	98.88%	0.6007	95.98%
5	0.1073	99.08%	0.6890	95.10%
6	0.1026	99.11%	0.3097	96.85%
7	0.0962	99.17%	0.5662	96.45%

Three observations stand out. First, learning was already strong after a single epoch. Second, the best development checkpoint did not occur at the final completed epoch, which indicates that loss on sampled episodes fluctuated even while the training curve remained smooth. Third, the generalization gap stayed moderate rather than widening sharply, which suggests that the embedding network learned more than a narrow memorization of training episodes.

The size of the improvement is easier to interpret when converted into relative changes. Training loss fell by about 91.5% between epochs 1 and 7, while training accuracy increased by 11.07 percentage points. On the development side, accuracy improved from 92.50% in epoch 1 to 96.85% at the best checkpoint in epoch 6, a gain of 4.35 percentage points. Development loss declined from 0.8702 to 0.3097 at the same checkpoint. These changes indicate that the detector did not only fit the training episodes more closely; it also improved its separation of bonafide and spoofed speech on new speaker episodes drawn from the development partition.

D. Training and Development Curves

The implementation report included a loss curve and a prototypical accuracy plot. They are referenced here as placeholders because the aim of this paper is to provide compile-safe IEEE source that can later incorporate the finalized graphics.



Fig. 4. Training and development loss across the seven completed epochs.

Fig. 4 is important because it shows that the training objective became small quickly, while Fig. 5 shows that the learned embedding also retained strong development performance. Used together, they suggest that the model learned a usable geometry and not only from reducing the auxiliary losses.

The shapes of the two curves also help separate optimization behavior from generalization behavior. Training loss decreased smoothly, which indicates stable optimization under the chosen learning rate, weight decay, and gradient clipping settings. Development loss fluctuated more because each evaluation pass used a fresh sample of episodes. Even so, the overall direction of the development curve remained favorable. This combination of smooth training optimization and noisy but

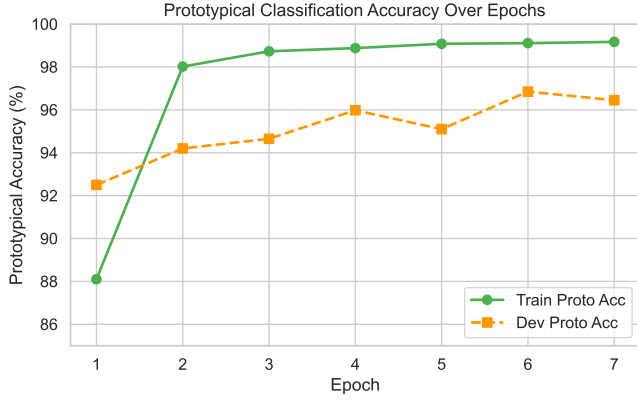


Fig. 5. Training and development prototypical classification accuracy across epochs.

improving development performance is consistent with a well-behaved episodic training process.

E. Dataset Composition and Checkpoint Behavior

The report also documented that the training partition is class-imbalanced, with bonafide speech making up roughly 10% of the samples and spoofed speech about 90%, distributed across the six known attack types. Episodic sampling mitigated this imbalance by drawing equal support and query counts per class during each episode. This design matters because ordinary minibatch training on the raw class proportions could have biased the detector toward spoof prevalence and away from a balanced decision rule.

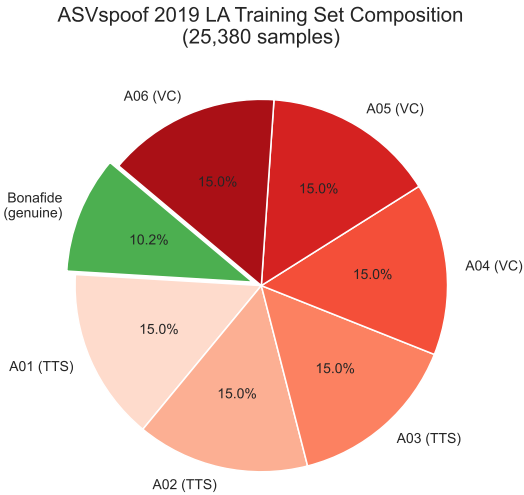


Fig. 6. Class and attack distribution in the ASVspoo 2019 LA training partition.

The best checkpoint occurred at epoch 6 instead of epoch 7. This is consistent with episodic validation, where the sampled evaluation episodes vary in difficulty. The important empirical

point is not the one-epoch fluctuation itself, but the fact that the best development accuracy remained high and the loss minimum was materially better than in earlier epochs.

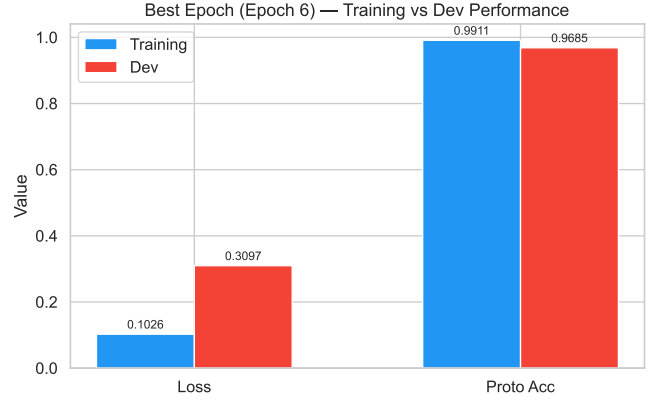


Fig. 7. Comparison of training and development performance at the best checkpoint.

F. Training Efficiency

Runtime is a result in its own right because it shapes whether the framework can support the rapid update cycle expected in fraud operations. Each epoch required about 44.2 minutes on the RTX 3090, or roughly 2,655 seconds, with only small variance across the seven runs. The main source of this cost was dual-stream self-supervised feature extraction inside every training episode.

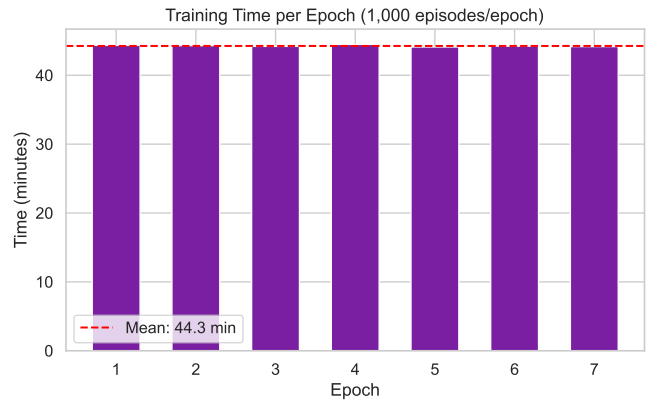


Fig. 8. Training time per epoch for the implemented system.

The convergence-rate figure described in the implementation report supports the same reading. The largest training-loss reduction occurred between epochs 1 and 2, and the marginal improvement became much smaller by epoch 7. This indicates that the system was approaching a plateau in the currently explored region of the parameter space, even though the planned training budget had not yet been exhausted.



Fig. 9. Per-epoch reduction in training loss, illustrating rapid early convergence and smaller later gains.

VII. RESULTS ANALYSIS

The first analytic conclusion is that the fused representation is effective on the known-attack setting that was actually evaluated. Reaching 88.10% training prototypical accuracy after a single epoch and then moving above 99% by epoch 5 indicates that the feature stack gave the embedding network a strong starting point. The improvement is too large to attribute only to easier episode sampling. The combination of self-supervised speech encoders and cepstral features appears to create a feature space in which bonafide and spoofed speech are already partly separable before substantial fine-tuning.

The second conclusion concerns generalization. The best development accuracy of 96.85% was obtained on speakers disjoint from training, so the detector was not merely overfitting speaker identity. The remaining gap between training and development performance, approximately 2-3 percentage points in accuracy at the best checkpoint, is consistent with mild overfitting but not with collapse. In security terms this matters because a model that depends strongly on speaker-specific cues is poorly suited to a bank, where the same fraud tool may be used against many customers whose voices were never seen during training.

The third conclusion concerns validation noise. Development loss did not decrease monotonically, and the final completed epoch was not the best checkpoint. This is not surprising in episodic learning, where evaluation is performed on sampled tasks rather than on a deterministic fixed batch order. The more informative pattern is the overall improvement in both development loss and accuracy between the first and best checkpoints. From epoch 1 to epoch 6, development loss dropped from 0.8702 to 0.3097, and development accuracy increased from 92.50% to 96.85%. Those changes are large enough to support the claim that the model learned a progressively stronger representation despite episodic variance.

The fourth conclusion is that the architecture is promising but expensive. The self-supervised feature path likely contributed materially to the strong convergence, which is consistent with recent literature on fused SSL features [17],

[20]. However, the same design also drove the 44.2-minute per-epoch cost. In a production bank, such training cost may be acceptable for offline model refreshes but less attractive for rapid incident response unless embeddings are cached or the front end is distilled into a smaller model. This trade-off should be stated plainly because detection quality and update speed are both operational requirements.

Class imbalance gives the accuracy values additional context. Since bonafide speech is the minority class in the raw training partition, a naive learner could drift toward spoof-heavy predictions if it were optimized on unbalanced mini-batches. That did not happen here because each episode enforced class balance. The strong development accuracy is therefore more informative than it would be under ordinary imbalanced batching. It indicates that the model learned a usable bonafide prototype, not only a strong spoof prototype.

Checkpoint behavior also matters from a model-governance perspective. The best development checkpoint appeared before the last completed epoch, and earlier epochs 1 and 2 were also retained when they achieved temporary minima in development loss. This is a sensible practice for security models, where a monitored validation signal should control model release more strongly than the final optimization step. In a banking deployment, the same principle would support conservative release policies and rollback options whenever a newly trained detector shows unstable validation behavior.

The fifth conclusion is more cautious. The study was motivated by few-shot continual learning, but the reported empirical evidence validates mainly the prerequisite stage: representation learning on known attacks. That stage is necessary because prototype adaptation will not help if the initial embedding is weak. The current results show that the embedding is strong enough to justify the next phase of experimentation. They do not yet prove that the full few-shot continual objective has been met. The research question is therefore narrowed, not resolved. The implementation has established a credible base model, not a finished deployment-ready countermeasure.

Comparison with prior work should be framed in the same spirit. The implementation report interpreted the 96.85% development accuracy as encouraging relative to traditional handcrafted-feature baselines and as consistent with literature showing the benefit of fused self-supervised speech features [17], [18]. That interpretation is reasonable, but it is not a substitute for direct reproduced baseline experiments. A controlled ablation study remains necessary to isolate the contribution of the SSL stream, the cepstral stream, the auxiliary attack head, and the EWC component.

From a financial perspective, the result that matters most is not the final decimal point on development accuracy. It is the shape of the evidence. The system learned quickly, retained strong development performance, and did so using a design that can in principle accept small support sets for new attacks. That combination is aligned with the threat model identified in the literature review. It suggests that the project has moved beyond a generic anti-spoofing experiment and toward a framework that speaks directly to operational fraud

response needs.

VIII. LIMITATIONS

The study has six concrete limitations.

- 1) Only seven of the planned 50 training epochs had completed when the implementation results were recorded. The reported numbers therefore describe an intermediate training state and not the final model ceiling.
- 2) The primary claim of few-shot adaptation to unseen attacks was not yet validated experimentally in the implementation report. The held-out attacks A07-A19 were reserved for later evaluation, so no reported Backward Transfer or post-adaptation retention figures are available in the present paper.
- 3) The implementation used a single benchmark, ASVspoof 2019 LA. Although that corpus is standard and useful, it contains studio-quality speech that does not fully reflect noisy financial call-center traffic, mixed devices, or conversational transaction language.
- 4) The preprocessing pipeline simulated codec conditions, but it did not close the entire realism gap identified in the literature. Effects such as cross-talk, network jitter, call-center room acoustics, and authentic fraud conversation structure remain outside the present experiments.
- 5) The dual-stream feature extractor is computationally heavy. With about 44.2 minutes per epoch and an estimated 2-3 seconds for a single 3-second segment during feature extraction on GPU, the current system is not yet optimized for low-latency deployment.
- 6) The paper includes literature-grounded discussion of fairness, partial fake attacks, and multimodal financial fraud integration, but the implementation does not yet evaluate those issues directly.

These limitations are not peripheral. They define the difference between a strong benchmark prototype and a bank-ready detection service. The practical value of the present work lies in making those boundaries explicit while still documenting the parts of the system that have been verified.

IX. CONCLUSION

This paper combined a finance-oriented literature review, a proposed system design, and an implementation report into a single study of audio deepfake detection for financial voice authentication. The synthesis showed that the field has made clear progress in detector architecture, self-supervised feature learning, and benchmark realism, yet still lacks a unified framework that addresses channel mismatch, unseen attacks, and forgetting in one design. To address that gap, the paper formulated the task as a few-shot continual learning problem and presented a dual-stream prototype-based detector with Elastic Weight Consolidation.

The implemented system instantiated that design on ASVspoof 2019 LA and produced strong intermediate results on known attacks: training loss fell from 1.1295 to 0.0962, training prototypical accuracy reached 99.17%, and best development accuracy reached 96.85%. These results indicate

that the learned embedding separates bonafide and spoofed speech effectively and provides a sound base for the next stage of experimentation. At the same time, the paper has shown why honest reporting matters: the full few-shot adaptation and retention claims remain to be tested on the held-out attacks. The contribution of the present work is therefore a grounded framework, a verified implementation core, and a clear path toward finance-relevant evaluation.

X. FUTURE WORK

Future work should proceed in four specific directions.

- 1) Complete the held-out attack study on A07-A19 and report adaptation accuracy, EER, precision, recall, F1 score, and Backward Transfer after each few-shot update stage. This is the direct empirical test of the continual learning claim.
- 2) Build or curate a finance-specific telephone corpus containing authentic transaction phrases, codec variation, device diversity, and partial-fake manipulations. Such a dataset would align evaluation with the banking threat model identified in the literature [4], [5], [10].
- 3) Reduce computational cost by caching frozen SSL embeddings, distilling the dual-stream front end into a smaller student model, or replacing repeated online extraction with a lighter feature server. This would make the framework more practical for rapid fraud-response updates.
- 4) Extend the detector into a multimodal fraud stack that combines audio evidence with transaction metadata, behavioral biometrics, and network-level anomaly signals. Financial fraud is inherently multi-signal, and audio should be treated as one part of a broader decision system [11], [12].

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